FLYING-WING STEADY RICCI SOLITONS AND NON-UNIQUENESS OF THE COLLAPSED CIGAR PRODUCT

ARGUS OPEN RESEARCH PROJECT

Abstract. AIMPL Problem 6.7 asks whether the cylinder product $\mathbb{R} \times \Sigma_{\text{cigar}}$ is the unique collapsed three-dimensional steady gradient Ricci soliton. Using Lai's $O(2)$-symmetric flying-wing theorem (2020) specialized to dimension $n = 3$ and asymptotic angle $\alpha = 1/2$, we verify that the resulting complete steady gradient Ricci soliton (M^3, g, f) has strictly positive sectional and Ricci curvature (hence is not isometric to $\mathbb{R} \times \Sigma_{\text{cigar}}$) and is collapsed at all scales under Lai's Definition 3.4 curvature-controlled spatial collapse convention.

1. Main Theorem

Let $\Sigma_{\text{cigar}} = (\mathbb{R}^2, \frac{dx^2+dy^2}{1+x^2+y^2})$ be Hamilton's cigar soliton. A complete Riemannian three-manifold (M^3, g) is *collapsed at all scales* in the sense of Lai (Definition 3.4) if there exists no $\kappa > 0$ for which (M^3, g) is κ-noncollapsed below all scales; equivalently, for every $\kappa > 0$ there exist a point $p \in M^3$ and a scale $r > 0$ such that $|\mathrm{Rm}| \le r^{-2}$ on $B_g(p, r)$ and $\mathrm{Vol}_g(B_g(p, r))/r^3 < \kappa$.

Theorem 1.1. *Let (M^3, g, f) be the complete three-dimensional $O(2)$-invariant steady gradient Ricci soliton with asymptotic cone angle $\alpha = 1/2 \in (0, \pi/2)$ constructed by Lai (Theorem 1.1 and Corollaries 1.3, 1.5). Then:*

(1) *(M^3, g, f) satisfies the steady gradient Ricci soliton equation $\mathrm{Ric}_g + \nabla^2 f = 0$ on the complete connected three-manifold $M^3 \cong \mathbb{R}^3$.*

(2) *$\mathrm{Ric}_g > 0$ everywhere on M^3, whereas $\mathbb{R} \times \Sigma_{\text{cigar}}$ has $\mathrm{Ric}(\partial_t, \partial_t) = 0$; hence (M^3, g) is not isometric to $\mathbb{R} \times \Sigma_{\text{cigar}}$.*

(3) *Along the two-dimensional wing asymptotic to $\Sigma_{\text{cigar}} \times S^1$ at finite S^1-fiber length, for every integer $j \ge 1$ there exist a point $p_j \in M^3$ and radius $r_j = j$ such that $|\mathrm{Rm}_g| \le r_j^{-2}$ on $B_g(p_j, r_j)$ and*

$$\frac{\mathrm{Vol}_g(B_g(p_j, r_j))}{r_j^3} < \frac{32}{j}.$$

Thus for every $\kappa > 0$, choosing $j > 32/\kappa$ witnesses that (M^3, g) is collapsed.

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